

Introduction to Probabilistic Machine Learning

Factor Graphs

Ralf Herbrich

Overview

1. Probabilistic Machine Learning
2. The Sum-Product Algorithm
3. Ranking Problems
4. TrueSkill

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*Unit 4 – Graphical Models:
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Probabilistic Machine Learning: Terminology

■ Given:

1. **Training Data:** $D \in (\mathcal{X} \times \mathcal{Y})^n$ of n (labelled) examples from the input space $\mathcal{X}$ and output space $\mathcal{Y}$
 - Binary Classification: $\mathcal{Y} = \{0,1\}$
 - Regression: $\mathcal{Y} = \mathbb{R}$
2. **Prior belief over functions from $\mathcal{X}$ to $\mathcal{Y}$:** $p(f), f \in \mathcal{F}$
 - Space of functions, $\mathcal{F}$, is also called *hypothesis space*.
3. **Likelihood of function:** $p(D|f) =: \ell(f)$
 - Link between data and functions
 - Normalizes over D but not over f – never say “likelihood of data”!
 - Models all assumptions how data/labels are generated from a function

■ Key Questions in Machine Learning:

- **Prediction:** What is $p(y|x, D)$ for an example x and having seen D ?
- **Decision Making:** What prediction $\hat{y}$ shall be made for an example x having seen D ?

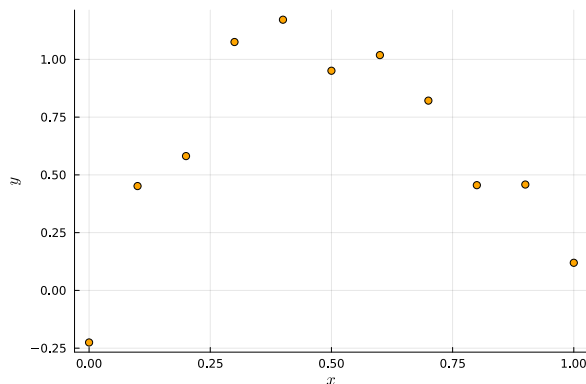
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Probabilistic Machine Learning: Polynomial Regression

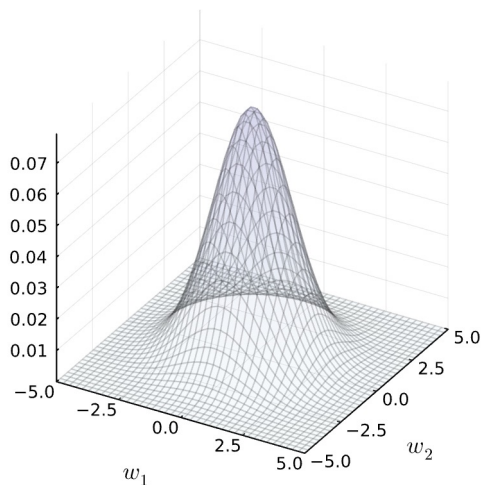
Training Data

$$D \in \mathbb{R}^2$$



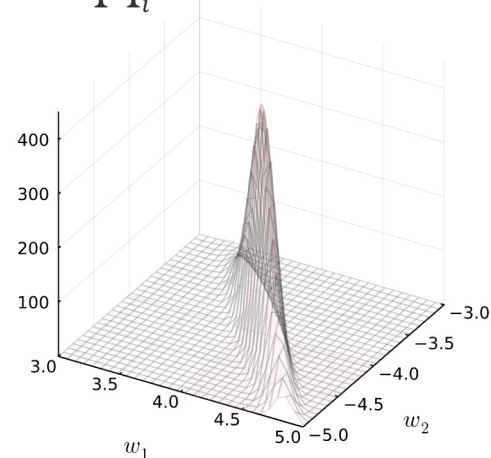
Prior

$$p(\mathbf{w}) = \mathcal{N}(w_1; 0, 1) \cdot \mathcal{N}(w_2; 0, 1)$$



Likelihood

$$\ell(\mathbf{w}) = \prod_i \mathcal{N}(y_i; w_1 x_i + w_2 x_i^2, \sigma^2)$$



$$f(x; \mathbf{w}) = w_1 \cdot x + w_2 \cdot x^2$$

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Predictions and Predictive Distributions

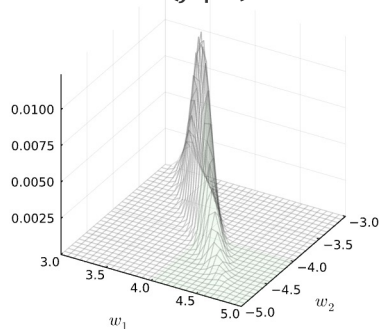
- Predictive distribution using the sum rule of probability

$$p(y|x, D) = \int p(y|x, f) \cdot p(f|D) df$$

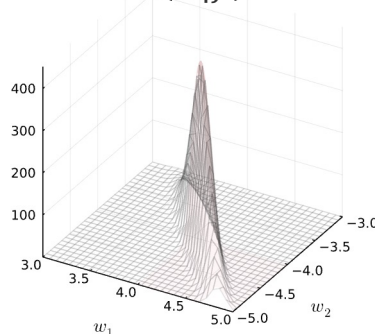
- But how do we get $p(f|D)$? Bayes' rule!

$$p(f|D) = \frac{p(D|f) \cdot p(f)}{p(D)}$$

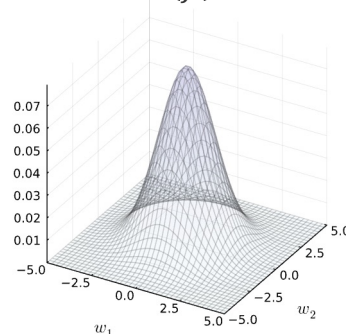
$P(f|D)$



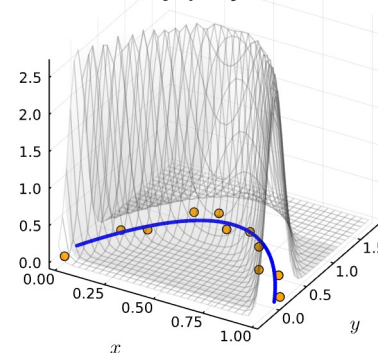
$P(D|f)$



$P(f)$



$P(y|x, f)$



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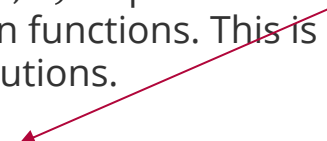
Probabilistic Machine Learning: Bayesian Inference

■ Two computational difficulties:

1. **Posterior** $p(f|D)$ requires the *multiplication* of likelihood with prior which often results in a distribution which is no longer in a family with very few parameters.

$$p(f|D) = \frac{p(D|f) \cdot p(f)}{p(D)} \propto \ell(f) \cdot p(f)$$


2. **Predictive distribution** $p(y|x, D)$ requires the *summation* of the data distribution over all prediction functions. This is only feasible for a small number of parametric distributions.

$$p(y|x, D) = \int p(y|x, f) \cdot p(f|D) df$$


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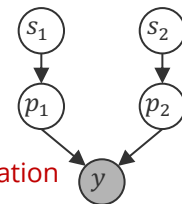
Graphical Models

■ Challenge: How to formulate complex likelihoods & priors for *actual* data?

□ **Example:** Match outcomes $y \in \{-1, 1\}$ (data) for a head-to-head match between two players

– **Prior:** $p(\mathbf{s}) = \mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2)$ ← skill belief

– **Likelihood:** $p(y|\mathbf{s}) = \int \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0) dp_1 dp_2$ ← marginalization



Match outcome

Player performance

■ Learning: In order to learn from data for most data models, we need to marginalize (“sum-out”) all non-observed variables given the observed variables (i.e., data).

■ Problem: Naïve summation scales exponentially because we have a sum of products (i.e., product of conditional distributions of all latent variables)!

□ **Example:** Consider an example of n Bernoulli variables $x_1, \dots, x_n$

$$p(x_1) = \sum_{x_2=0}^1 \sum_{x_3=0}^1 \dots \sum_{x_n=0}^1 p(x_1, x_2, \dots, x_n) \quad \leftarrow 2^{n-1} \text{ summations}$$

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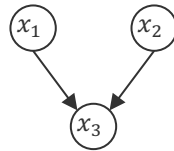
Factor Graphs



Brendan Frey
(1968 –)

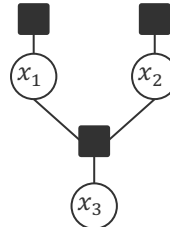
- **Factor Graph (Frey, 1998).** Given a product of m functions $f_1, f_2, \dots, f_m$, each over a subset of n variables $x_1, x_2, \dots, x_n$, a factor graph is a bipartite graphical model with m factor nodes and n variable nodes where an undirected edge connects f_i and x_j if and only if the function f_i depends on x_j .

Bayesian network



$$p(x_1, x_2, x_3) = p(x_1) \cdot p(x_2) \cdot p(x_3 | x_1, x_2)$$

Corresponding factor graph

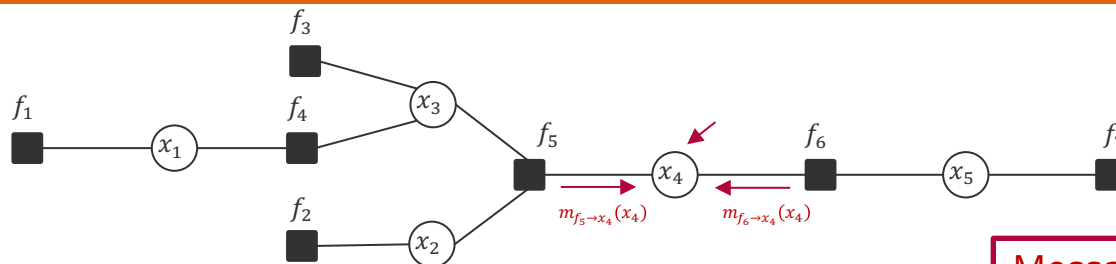


$$p(x_1, x_2, x_3) = f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3, x_1, x_2)$$

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Sum-Product Algorithm: Marginals



Message $m_{f_j \rightarrow x_i}(x_i)$ is the sum over all variables in the subtree rooted at f_j

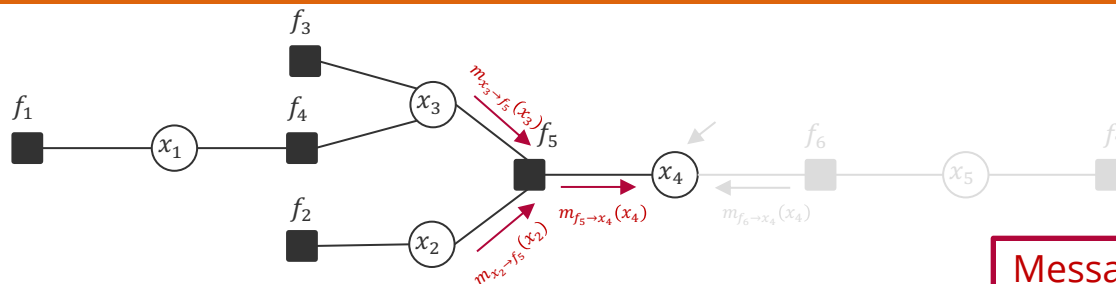
$$\begin{aligned}
 p(x_4) &= \sum_{\{x_1\}} \sum_{\{x_2\}} \sum_{\{x_3\}} \sum_{\{x_5\}} f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3) \cdot f_4(x_1, x_3) \cdot f_5(x_2, x_3, x_4) \cdot f_6(x_4, x_5) \cdot f_7(x_5) \\
 &= \left[\sum_{\{x_1\}} \sum_{\{x_2\}} \sum_{\{x_3\}} f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3) \cdot f_4(x_1, x_3) \cdot f_5(x_2, x_3, x_4) \right] \cdot \left[\sum_{\{x_5\}} f_6(x_4, x_5) \cdot f_7(x_5) \right] \\
 &\quad m_{f_5 \rightarrow x_4}(x_4) \qquad \qquad \qquad m_{f_6 \rightarrow x_4}(x_4)
 \end{aligned}$$

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Marginals are the product of all incoming messages from neighbouring factors!

Sum-Product Algorithm: Message from Factor to Variable



Message $m_{x_i \rightarrow f_j}(x_i)$ is the sum over all variables in the subtree rooted at x_i

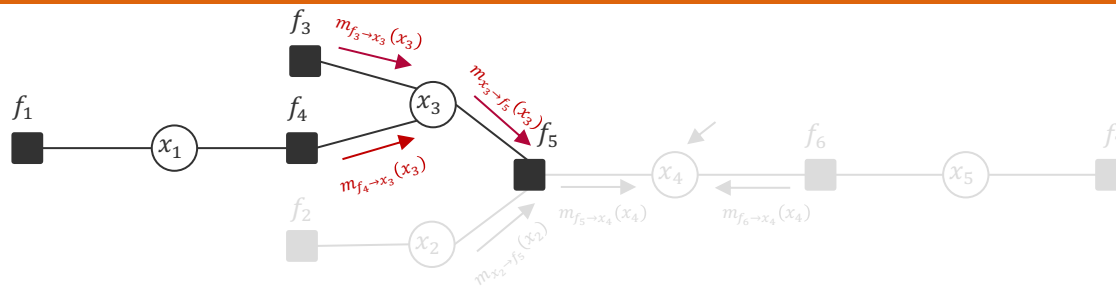
$$\begin{aligned}
 m_{f_5 \rightarrow x_4}(x_4) &= \sum_{\{x_1\}} \sum_{\{x_2\}} \sum_{\{x_3\}} f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3) \cdot f_4(x_1, x_3) \cdot f_5(x_2, x_3, x_4) \\
 &= \sum_{\{x_2\}} \sum_{\{x_3\}} f_5(x_2, x_3, x_4) \cdot \underbrace{[f_2(x_2)]}_{m_{x_2 \rightarrow f_5}(x_2)} \cdot \underbrace{\left[\sum_{\{x_1\}} f_1(x_1) \cdot f_3(x_3) \cdot f_4(x_1, x_3) \right]}_{m_{x_3 \rightarrow f_5}(x_3)}
 \end{aligned}$$

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Messages from a factor to a variable sum out all neighboring variables weighted by their incoming message

Sum-Product Algorithm: Message from Variable to Factor



$$\begin{aligned}
 m_{x_3 \rightarrow f_5}(x_3) &= \sum_{\{x_1\}} f_1(x_1) \cdot f_3(x_3) \cdot f_4(x_1, x_3) \\
 &= [f_3(x_3)] \cdot \left[\sum_{\{x_1\}} f_1(x_1) \cdot f_4(x_1, x_3) \right] \\
 &\quad m_{f_3 \rightarrow x_3}(x_3) \quad m_{f_4 \rightarrow x_3}(x_3)
 \end{aligned}$$

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Messages from a variable to a factor multiply incoming message from neighboring factors

Sum-Product Algorithm



Robert McEliece
(1942 – 2019)

- **Sum-Product Algorithm (Aji-McEliece, 1997).** Putting it all together, we have

$$\begin{aligned} p(x) &= \prod_{f \in \text{ne}(x)} m_{f \rightarrow x}(x) \\ m_{f \rightarrow x}(x) &= \sum_{\{x' \in \text{ne}(f) \setminus \{x\}\}} \cdots \sum_{\{x'' \in \text{ne}(f) \setminus \{x\}\}} f(x, x', \dots, x'') \prod_{x' \in \text{ne}(f) \setminus \{x\}} m_{x' \rightarrow f}(x') \\ m_{x \rightarrow f}(x) &= \prod_{f' \in \text{ne}(x) \setminus \{f\}} m_{f' \rightarrow x}(x) \end{aligned}$$

- **Basis:** Generalized distributive law (which also holds for max-product)
- **Efficiency:** By storing messages, we
 - Only have to compute local summations in $O(2^T)$ where degree $T = \max_f |\text{ne}(f)|$!
 - All marginals can be computed recursively in $O(E \cdot 2^T)$ vs $O(2^n)$ (where E is the number of edges of the factor graph)!

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Even more efficiency

- **Redundancies.** By the very definition of messages and marginals

$$p(x) = \prod_{f \in \text{ne}(x)} m_{f \rightarrow x}(x) = m_{f' \rightarrow x}(x) \cdot \prod_{f \in \text{ne}(x) \setminus \{f'\}} m_{f \rightarrow x}(x) \quad \leftarrow m_{x \rightarrow f'}(x)$$

- **Interpretation.** Application of Bayes' rule at a variable x at factor f

$$p(x) = m_{f \rightarrow x}(x) \cdot m_{x \rightarrow f}(x)$$

posterior Likelihood × normalization prior

- **Storage Efficiency.** We only store the marginals $p(x)$ and $m_{f \rightarrow x}(x)$ because

$$m_{x \rightarrow f}(x) = \frac{p(x)}{m_{f \rightarrow x}(x)}$$

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The Skill Rating Problem

■ Given:

- **Match outcomes:** Orderings among k teams consisting of $n_1, n_2, \dots, n_k$ players.

Team		Score				
1st	Red Team	50				
2nd	Blue Team	40				
	Level	Gamertag	Avg. Life	Best Spree	Score	
1st	10	BlueBot	00:00:49	6	15	
1st	7	SniperEye	00:00:41	4	14	
1st	9	ProThePirate	00:01:07	3	13	
1st	10	dazdemon	00:00:59	3	8	
2nd	10	WastedHarry	00:00:41	4	17	
2nd	3	Ascla	00:00:37	2	10	
2nd	9	Antidote4Losing	00:00:41	2	9	
2nd	12	Blackknight9	00:00:48	3	4	

	Level	Gamertag	Avg. Life	Best Spree	Score	
1st	N/A	SniperEye	N/A	N/A	25	
2nd	N/A	xXxHALOxXx	N/A	N/A	24	
3rd	N/A	AjaySandhu	N/A	N/A	15	
3rd	N/A	AjaySandhu(G)	N/A	N/A	15	
5th	N/A	Robert115	N/A	N/A	11	
5th	N/A	TurboNegro84(G)	N/A	N/A	11	
7th	N/A	TurboNegro84	N/A	N/A	5	
8th	N/A	SniperEye(G)	N/A	N/A	1	

■ Questions:

1. Skill s_i for each player such that $s_i > s_j \Leftrightarrow P(\text{Player } i \text{ wins}) > P(\text{Player } j \text{ wins})$
2. Global ranking among all players
3. Fair matches between teams of players

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Two-Player Match Outcome Model

- **Simple Two-Player Games:** Our data is the identity i and j of the two players and the outcome $y \in \{-1, +1\}$ of a match between them

- **Bradley-Terry Model (1952):** Model of a win of player i given skills s_i and s_j is

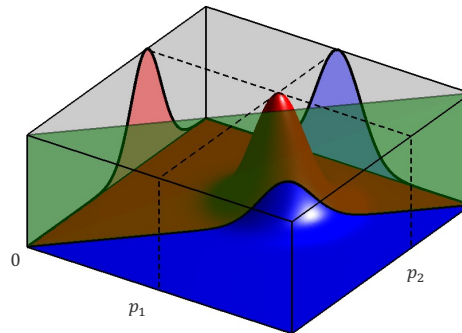
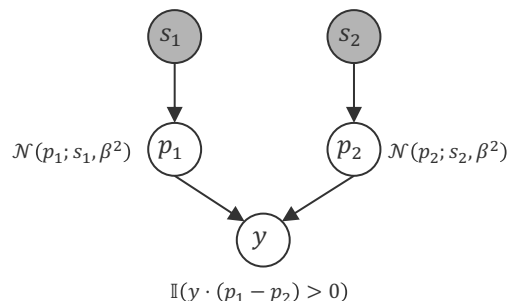
$$P(y = 1 | s_i, s_j) = \frac{\exp(s_i)}{\exp(s_i) + \exp(s_j)} = \frac{\exp(s_i - s_j)}{1 + \exp(s_i - s_j)}$$

Logistic sigmoid in skill difference

- **Thurstone Case V Model (1927):** Model of a win given skills s_i and s_j is

$$P(y = 1 | s_i, s_j) = \int_0^\infty \mathcal{N}(t; s_i - s_j, 2\beta^2) dt$$

Probit sigmoid in skill difference



Ralph A. Bradley
(1923 – 2001)



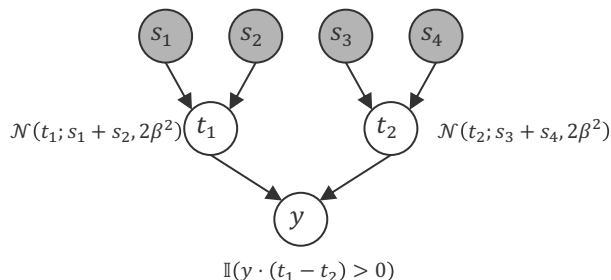
Louis Leon Thurstone
(1887 – 1955)

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Two-Team Match Outcome Model

- **Team Assumption:** Performance of a team is the sum of the performances of its players



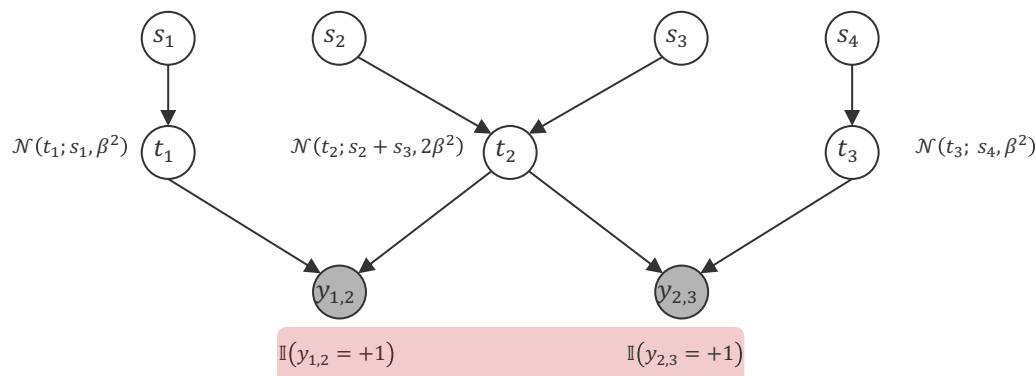
- **Pro:** Games where the team scores are additive (e.g., kill count in first-person shooter)
- **Con:** Games where the outcome is determined by a single player (e.g., fastest car in a race)
- **Observation:** Match outcomes correlate the skills of players
 - **Same Team:** Anti-correlated
 - **Opposite Teams:** Correlated

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From Match Outcomes to Pairwise Rankings

- **Learning:** In the ranking setting, we observe multi-team match outcomes and want to infer the skills of all single players!
- **Idea:** Leverage the transitivity of the real line of latent scores!



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By transitivity, this implies $t_1 > t_3$

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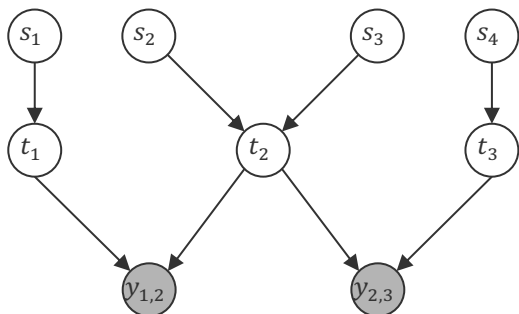
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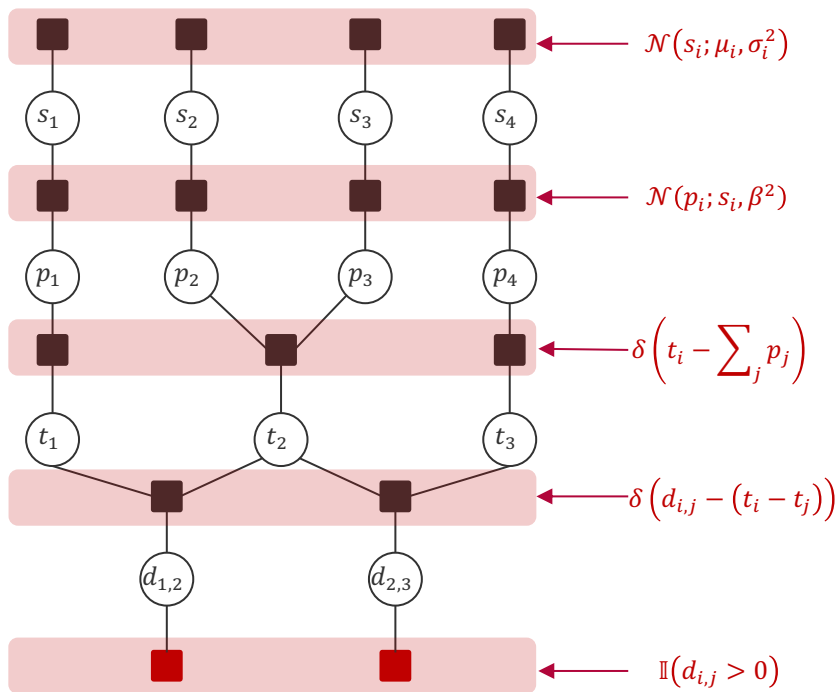
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TrueSkill Factor Graphs

Bayesian Network



Factor Graph

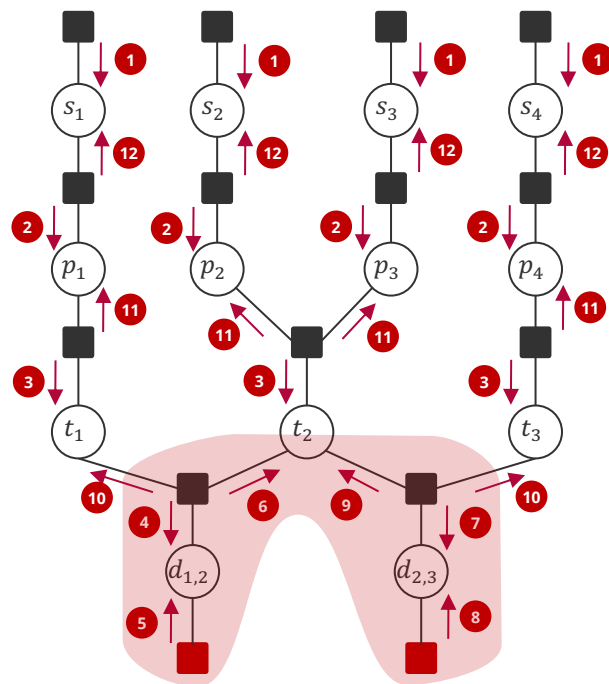


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(Approximate) Message Passing in TrueSkill Factor Graphs

TrueSkill Factor Graph



$$\mathcal{N}(s_i; \mu_i, \sigma_i^2)$$

$$\mathcal{N}(p_i; s_i, \beta^2)$$

$$\delta\left(t_i - \sum_j p_j\right)$$

$$\delta\left(d_{i,j} - (t_i - t_j)\right)$$

$$\mathbb{I}(d_{i,j} > 0)$$

Four Phases

1. Pass prior messages (1)
2. Pass messages *down* to the team performances (2 to 3)
3. Iterate the approximate messages on the pairwise team differences (4 to 9)
4. Pass messages back from *up* from team performances to player skill (10 – 12)

Since this is a *tree*, the algorithm is guaranteed to converge!

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Decision Making: Match Quality and Leaderboards

■ Match Quality: Decide if two players i and j should be matched

- **Idea:** Pick the pair (i, j) where the two players have equal skills

$$\text{Quality}(i, j) = \frac{P(p_i \approx p_j | \mu_i - \mu_j, \sigma_i^2 + \sigma_j^2)}{P(p_i \approx p_j | \mu_i - \mu_j = 0, \sigma_i^2 + \sigma_j^2 = 0)}$$

- **Observation:** This pair (i, j) approximately maximizes the information (entropy!) of the predicted match outcome because it gets closest to 50% winning probability

■ Leaderboard: Decide how to display the best to worst player

- **Observation:** There is an asymmetry in making a ranking mistake
 - **Cheap:** Ranking a truly good player lower than they should be (why?)
 - **Expensive:** Ranking a truly bad player higher than they should be (why?)
 - The loss minimizer of this decision process is a **quantile** $\mu - k \cdot \sigma$ with $k > 0$



1	27	SEWICSYDE OWNS
2	26	FATAL REVENGE
3	25	Paranoia 1
4	25	Paulk
5	25	lxX OMG Xxl
6	25	BittyTom
7	24	brian 2007
8	24	SEXY MOZES
9	24	droplates
10	24	jaCKdaSaMuRai
11	24	Il Me Il
12	24	iamNightMare
13	24	a retarded007
14	24	Perfected Brit
15	24	THE MUFFIN MANx
16	23	TheVunit
17	23	Mr Sushi87

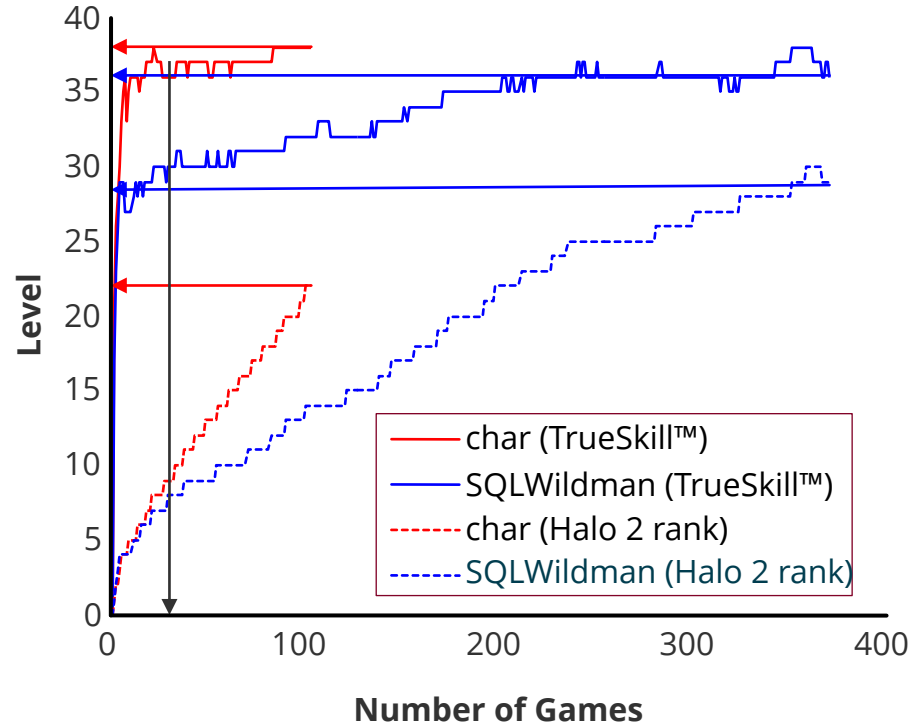
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Experimental Results

■ Data Set: Halo 2 Beta

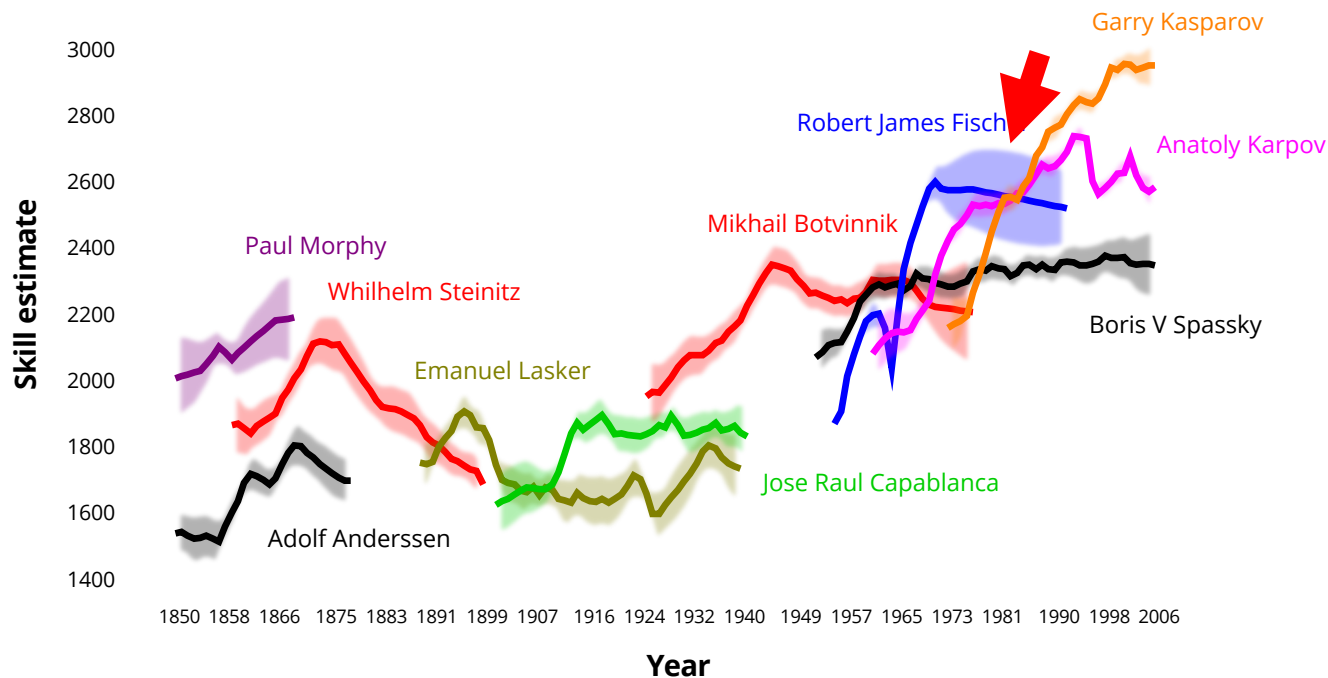
- 3 game modes
 - Free-for-All
 - Two Teams
 - 1 vs. 1
- > 60,000 match outcomes
- ≈ 6,000 players
- 6 weeks of game play
- Publically available



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TrueSkill-Through-Time: Chess Players



History of Chess
3.5M match outcomes
20 million variables
40 million factors

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See you next week!